

30/7/26

ITA0502 - Activity

Scenario Based Activity

① Salt and pepper noise with CCTV image:

Technique: Median filter

⇒ It removes salt and pepper noise effectively.

⇒ It preserves image edges better than averaging filters.

⇒ It replaces noisy pixels with the median value.

② Satellite image with Gaussian noise

Technique: Gaussian filter

⇒ It reduces gaussian noise by smoothing the image.

⇒ It uses a Gaussian kernel for filtering.

⇒ It improves overall image quality.

③ Medical x-ray with random noise:

Technique: Bilateral Filter

⇒ It removes random noise effectively

⇒ It preserves fine edges and important details

⇒ It is widely used in medical image processing

④ Scanned document with black and white dots:

Technique: Median filter

⇒ Removes random noise ⇒ Eliminates black

⇒ Removes and white dots. keeps text clear and readable. ideal for scanned documents.

⑤. Low - light image with heavy grain

Technique: Non-Local means filter

⇒ Reduces heavy grain noise. Preserves textures and details. Improves low-light images.

⑥. Foggy road image with low contrast

Technique: Histogram Equalization

⇒ Enhances image contrast.

⇒ Improves visibility in fog. Makes road features clearer.

⑦. Underwater image appears dull

Technique: Contrast stretching

⇒ It increases brightness and contrast.

⇒ It makes the image look clearer.

⑧. Chest x-ray with poor visibility

Technique: Adaptive histogram Equalization (CLAHE)

⇒ It improves local contrast

⇒ It reveals internal structures clearly.

⑨. Night-time traffic image:

Technique: Contrast stretching

⇒ It increases intensity variation

⇒ It makes objects more visible.

⑩. Grayscale image with narrow intensity range

Technique: Histogram Equalization

⇒ It spreads the intensity values.

⇒ It enhances overall image contrast.